

# SD FC Pricing Intelligence Platform

Executive Summary | VP of Data Analytics & Strategy

**28,064**  
2025 Avg Attendance

**8-14%**  
Forecast Error (MAPE)

**~\$207K**  
2025 Revenue Left on Table

**9-14%**  
Projected Uplift (Balanced)

## The Problem

San Diego FC completed its inaugural 2025 season averaging 28,064 fans per game in a 35,000-seat Snapdragon Stadium. Ticket prices were set once before the season using gut feel and competitor benchmarks. The secondary market told a different story: Baja Cup matches against Club Tijuana cleared at 40-58% above face value, while midweek games against weaker opponents sat below face. An estimated \$207,000 in revenue was left on the table in Year 1 and that number grows as the market matures.

## The Solution

A machine learning platform that recommends optimal ticket prices for every section and every game, grounded in what fans are actually willing to pay. It combines demand forecasting (XGBoost, 8-14% error rate), causal price elasticity (Double Machine Learning), and airline-style yield optimization (EMSR-b) calibrated against verified 2025 attendance data.

The platform outputs three pricing scenarios per section per game:

- **Conservative:** minimal price movement, targets 90%+ fill rate, lowest risk
- **Balanced (recommended):** optimizes revenue while protecting season ticket holder resale margins
- **Aggressive:** maximum revenue extraction for confirmed high-demand games only

Every recommendation includes a plain-English explanation: "Section 111 is \$8 below optimal. Drivers: rivalry game (+\$8), high secondary demand (+\$5), Saturday night (+\$3)."

## What Makes It Different

- **Causal, not correlated** -- isolates the true effect of price on demand, controlling for opponent quality, weather, and timing
- **Explainable** -- every recommendation comes with reasons your ticketing team can understand and challenge
- **Guardrailed** -- STH resale margins protected by design (primary never exceeds secondary minus 10%); any increase above 25% flagged for manual review
- **Three choices, not one** -- management picks the risk level, the model handles the math

## Implementation Path

| Phase   | Timeline    | What Happens                                                                                                |
|---------|-------------|-------------------------------------------------------------------------------------------------------------|
| Phase 1 | Now         | Offline recommendations. Run model before each pricing window, present scenarios, manually update Archtics. |
| Phase 2 | 3-6 months  | API integration with Ticketmaster Host for automated price updates with human approval gate.                |
| Phase 3 | 6-12 months | Real-time feedback loop with live sell-through tracking and mid-season recalibration.                       |

## First 90 Days

**Days 1-30:** Replace synthetic data with real Archtics/CRM feeds. Validate model accuracy on real transaction data.

**Days 31-60:** First live pricing recommendations for early-season home games. A/B test: control vs model-priced section groups. Dashboard live for ticketing team.

**Days 61-90:** First measurable results. Revenue uplift report. STH health validation. Phase 2 roadmap presented.

Built with Python, XGBoost, DuckDB, Streamlit, FastAPI, SHAP | Data: FBref, SeatGeek, StubHub, Vivid Seats, OddsPortal, MLS Official, Ticketmaster